

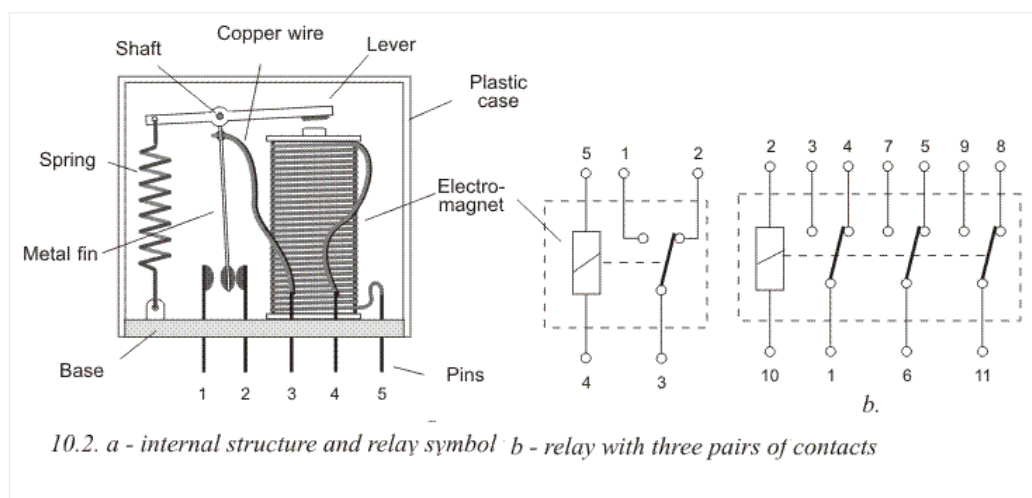


10.1 RELAYS

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A relay is an electro mechanical device which is commonly used to connect two different circuits. It can connect a low voltage circuit to a high voltage circuit or a low current circuit to a high current circuit or simply to isolate two circuits.

The simplest relay has one set of contacts (commonly called “change-over” contacts). Inside the relay is a coil (called a solenoid) and when the coil is energised, the centre core of the solenoid becomes magnetised and moves an arm closer to the coil. A “contact” is connected to this arm and the contact touches another contact to complete a circuit. The contacts are labeled “common” for the moving contact, “normally open” and “normally closed.” This can be seen in diagram 10.2 a:



A relay can be connected as the collector load of a transistor, as shown on 10.3. When sufficient collector current flows in the transistor, the relay is activated and any device connected to the contacts will be operational.

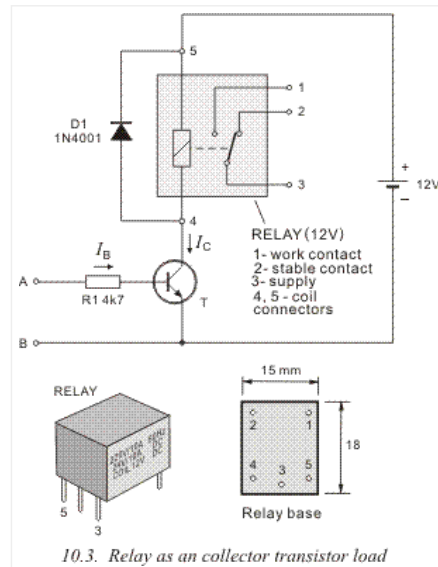
Since a relay is an electro mechanic component which is consisted of moving parts, it has a limited operational life span, and cannot be used for rapid switching. It would not be very effective using it in a, for example, light show which has frequent switching frequency (several hundreds or thousands times per hour). Each opening and closing of the contact is followed by sparks which would dramatically shorten the life of such device.

Coil values are “input values” or voltage and resistance values at which relay draws the lever and switches.

Usual coil voltages are 3V, 5V, 6V, 12V and 24V. They can be found printed on the relay's housing. These are all DC voltages, but there are AC voltage designed relays with 230V/250V. The current taken by the relay depends on the resistance of the coil. The coil resistance can be measured with a multimeter.

Current flowing through the coil is calculated using Ohm's law, by dividing the relay's voltage by its resistance. For example a 12v relay has a coil resistance of 300 Ohm, which means the current flow is:

$$I = U/R = 12/300 = 40\text{mA}.$$



2. Voltage on relay's contacts, also marked on the housing, is the maximum value allowed. Over-voltage will cause sparks inside the relay and possibly damage the contacts.

The maximum current rating for a relay is marked on the housing with all the other information. It is usually higher than 1A.



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